Mobility Solution: Four-Wheel Differential Drive on a Single-Pivot Rocker

I. Why Wheels

Europa's surface is fractured, chaotic terrain sitting at 90 K under 1.314 m/s^2 of gravity, and the mission asks the rover to build a spatially continuous subsurface profile rather than to visit a handful of points. That second part is what decides the mobility architecture. REQ.03.01 needs a radar trace placed every 100 m or better along a single along-track transect, and REQ.06.01 puts a sounding stop at every one of those intervals. A hopper or a legged walker gives you discrete samples with a discontinuity between each one, and every discontinuity is a place where the localization solution has to be re-established from nothing. Wheels give a continuous odometry signal, which is the thing the whole ±0.5 m localization budget in REQ.02.01 is built on.

Low gravity argues the other way, at least at first. A hopper is efficient at 1.3 m/s^2 and legs handle fractured blocks better than a rigid wheel does. Both lose on the same point: they are impulsive. A hop that ends badly on a 90 K ice block ends the mission, and there is no way to hand the problem to Earth inside the 44 min one-way delay. Wheels on a rocker fail gradually instead, which matters more than efficiency when the recovery loop is that long.

II. Configuration

Four independently driven wheels on a single-pivot rocker. No steering actuators. The chassis is the 0.90 m × 0.60 m equipment deck already carried in the mass budget, held 0.35 m above the surface so the belly-mounted sounder keeps its standoff, with a 0.70 m wheelbase and a 0.80 m track. The four tapered landing struts are jettisoned after checkout, so nothing but the wheels touches the ice during the survey.

a. **Wheels.** 0.32 m diameter, 0.20 m wide, rigid aluminum rim with 24 hardened cleats. There is no tire and no elastomer anywhere in the contact path, because nothing compliant survives a 90 K soak and repeated thermal cycling.

b. **Suspension.** A single rocker pivot on the chassis belly centerline with bent arms and elbow joints out to each hub. This is a rocker, not a rocker-bogie. Six wheels and two bogies would climb better, but they cost two more actuators, two more thermal isolation paths, and two more line items in a mass budget that is already at 136.9 kg against a 147.25 kg cap.

c. **Steering.** Differential drive. Opposed wheel rates rotate the chassis about the rocker pivot, which means every heading change is a deliberate skid. I will come back to why that is acceptable, since it is the least obvious choice in the design.

d. **Speed.** 2 cm/s commanded, driven in arcs no longer than 5 m.

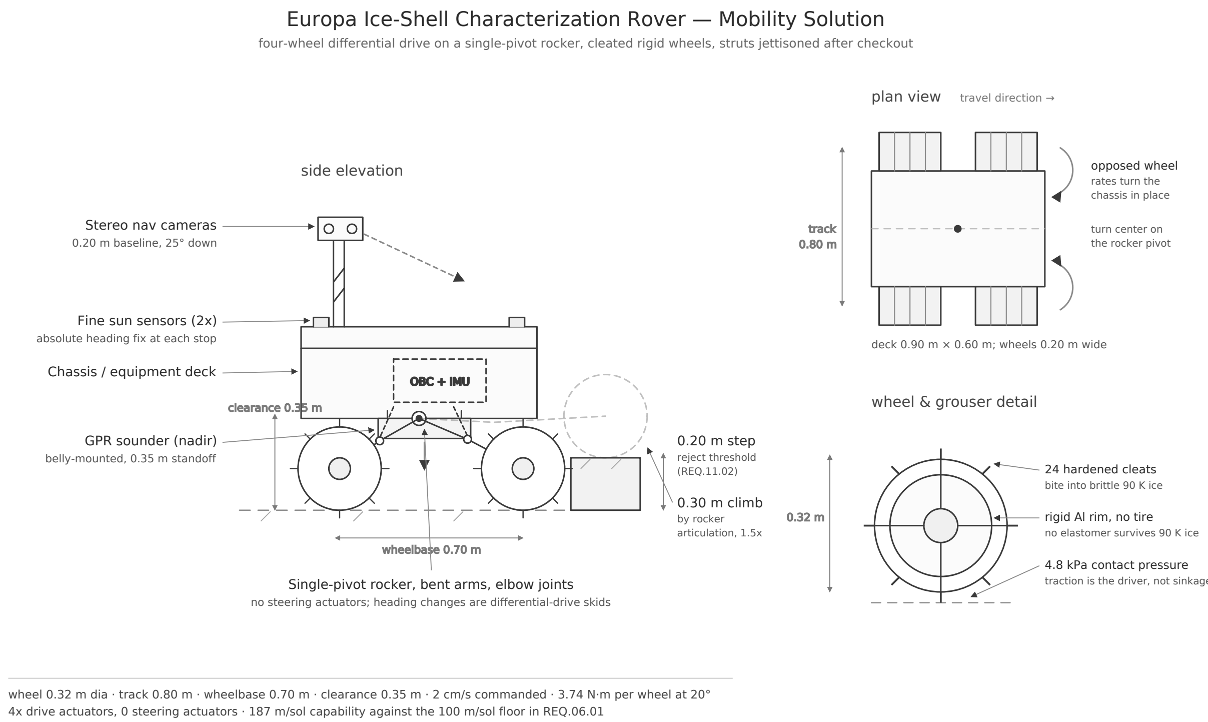

Figure 1: Mobility solution: side elevation, plan view, and wheel detail.

III. Traction and Load

Take the mass budget at its contingency cap of 147.25 kg. On Europa that weighs $147.25 \times 1.314 = 193.5\,\mathrm{N}$, or 48.4 N per wheel. Against a contact patch of roughly $0.20\,\mathrm{m} \times 0.05\,\mathrm{m}$ that is 4.8 kPa. Water ice has a compressive strength in the megapascals, so sinkage is not the problem here and there is no point sizing the wheels for flotation the way a Mars design has to. What the wheels have to do is generate shear.

Tractive force needed to hold the 20° slope limit that REQ.03.02 already imposes, with a rolling resistance coefficient of 0.15:

$$F = W\,(\sin 20^\circ + 0.15\cos 20^\circ) = 193.5 \times (0.342 + 0.141) = 93.4\,\mathrm{N},$$

which is 23.4 N per wheel, or 3.74 N m at a 0.16 m radius. The useful mechanical power at 2 cm/s is only 1.87 W. The 84 W peak already allocated to the four drive motors in the power budget is not going into moving the rover, it is going into breakout torque and into gearbox drag at 90 K, where the lubricant is stiff enough to dominate everything else. That is also why REQ.09.02 preheats the actuators above $-40\,^\circ\mathrm{C}$ before any commanded motion.

The available shear is the part that turns out to be favorable, and for a reason that is specific to this destination. Ice is slippery near its melting point because a thin meltwater film forms under contact pressure. At 90 K there is no film, and the friction coefficient for a hard slider on ice climbs toward 0.5 to 0.7 rather than the 0.05 you would get on a temperate glacier [1]. Taking 0.5 conservatively gives a limiting slope of 26.6° against a required 20°, so the cleats have margin without needing an aggressive grouser height. Europa ice behaves less like a skating rink and more like a brittle rock, which is good for traction and bad for the wheels, and it is the reason the cleats are hardened rather than machined into the rim.

Per-sol distance closes with room to spare. Four hours of drive time at 2 cm/s with a 65 % duty cycle after subtracting stops for imaging and hazard assessment gives 187 m, against the 100 m floor in REQ.06.01. The average rate the requirement actually demands is 6.9 mm/s, so the drive train is not the binding constraint on traverse rate. Hazard rejection and the sounding cadence are.

IV. Steering Without Steering Actuators

Skid steering on hard ice slips badly, and I want to be direct about that rather than hide it. A differential turn on a surface with $\mu \approx 0.5$ and no compliance in the contact patch produces wheel rotation that does not correspond to chassis rotation, so wheel odometry cannot be trusted to close a turn. The design accepts this and stops using wheel odometry as a position source. It becomes a slip observable instead: the residual between what the encoders report and what the stereo visual odometry measures is the slip estimate, which is the same approach the MER rovers used on Martian sand [2].

That decision only works because heading is fixed absolutely rather than integrated. A tactical-grade MEMS IMU holding 1.5 °/h of bias stability drifts 1.04° across a full 50 m segment at 2 cm/s, which alone would be 0.91 m of cross-track error and would blow the ±0.5 m budget in REQ.02.01 twice over. Breaking the segment into 5 m arcs and taking a sun-sensor heading fix at the end of each one caps the inertial contribution at 0.104°, and combined with the ±0.25° fix accuracy the heading error lands at 0.271°. Over 50 m that is 0.24 m of cross-track error. The full budget then closes:

Error source over a 50 m **segment**	**Contribution** (m)
Visual odometry scale and feature residual	0.30
Heading knowledge (0.271°)	0.24
Wheel slip on cleated water ice	0.20
Inter-agent range residual	0.15
Root-sum-square	**0.46**
REQ.02.01 limit	0.50

Table 1: Localization error budget for one 50 m traverse segment.

Eight percent of margin is thin, and slip is the term I would attack first if it came in worse than assumed, since it is the only one that responds to a mobility change rather than a sensor change. Widening the wheels or adding grouser height buys shear directly. Trading the differential turn for four steering actuators would buy more, and it would cost roughly 6 kg and four more cold-start mechanisms, so it stays on the shelf unless analog testing shows the slip term above 0.30 m.

V. Alignment With the Mission

The mobility design exists to serve three things the mission already committed to. It keeps the sounder within ±5° of nadir on uneven ice, because a single rocker pivot with ±0.5° of tilt knowledge gives 10 times the margin REQ.03.02 needs. It produces the continuous, georeferenced along-track profile REQ.03.01 asks for, at a 0.46 m localization error that fits inside REQ.02.01. And it holds 187 m of per-sol capability against a 100 m requirement, which leaves the schedule room to absorb the navigation holds REQ.11.04 will occasionally force.

The rocker also degrades in the right direction. Losing one drive actuator leaves three wheels and a chassis that can still be dragged along a reduced-rate traverse, where losing a steering actuator on a four-corner steered design leaves a wheel locked at an angle and fights every subsequent arc. Redundancy through independent failure modes is worth more here than performance at nominal, given that nobody is coming to fix it.

VI. Open Items

Three things are not settled. Cleat wear against brittle ice over a 30-sol traverse has no flight precedent, and abrasion at 90 K could change the effective friction coefficient in either direction. The 0.15 rolling resistance coefficient is borrowed from Martian regolith rather than measured on ice analog, so the 3.74 N m torque number should be treated as an estimate until a bench test replaces it. And the mass and power budgets do not yet carry mobility line items at all, which is flagged separately along with the 17.1 kg and 3.5 W that this design adds.

References:

References

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AI disclosure: I used AI to check the arithmetic behind the traction, torque, and localization error budget numbers, and to cross-check the low-temperature ice friction values against published measurements. I picked the mobility architecture, the wheel and suspension configuration, and the decision to run differential drive without steering actuators myself.